

User and Group Administration

Objectives

After completing this lesson, you should be able to:

- Explain user and group implementation
- Describe user and group configuration files
- Configure users and groups by using command-line utilities
- Implement user private groups (UPG)
- Configure password aging and the hashing algorithm
- Use the User Manager GUI tool
- Manage the use of `su` and `sudo` commands
- Describe user and group administration in Oracle Cloud Infrastructure
- Describe user and group implementation in the enterprise



Introduction to Users and Groups

- User account information is stored in `/etc/passwd`.
- Group information
 - Group information is stored in `/etc/group`.
 - Each user has a private group (UPG).
 - Users can belong to more than one group.
- Oracle Linux uses shadow passwords.
 - `/etc/shadow`: Hashed user passwords
 - `/etc/gshadow`: Hashed group passwords
 - `/etc/login.defs`: Security policies



User and Group Configuration Files

- Contents of `/etc/passwd`:
 - username: placeholder: UID: GID: GECOS: home dir: shell
- Contents of `/etc/shadow`:
 - username: hashed password: password aging information
- Contents of `/etc/group`:
 - groupname: placeholder: GID: comma-separated members
- Contents of `/etc/gshadow`:
 - groupname: hashed password: GID: comma-separated administrators: comma-separated members
 - Group passwords are rarely used.

Adding a User Account

- Use the `useradd` command to add a user:

```
# useradd [options] user_name
```

- Use the `passwd` command to create a password:

```
# passwd [options] user_name
```

- User default settings are stored in:
 - `/etc/default/useradd`
- Use the `-D` option to display or modify defaults:

```
# useradd -D [options]
```

- A new user's home directory is populated with files from:
 - `/etc/skel` directory
- To create a `nologin` user:

```
# useradd -s /sbin/nologin user_name
```

Modifying or Deleting User Accounts

- Use the `usermod` command to modify a user:

```
# usermod [options] user_name
```

- Example: To add a user to a secondary group (GID=1017):

```
# usermod -aG 1017 user_name
```

- Use the `userdel` command to delete a user:

```
# userdel [options] user_name
```

- Options to `userdel` include:
 - `-f`: Force removal even if the user is logged in.
 - `-r`: Remove the user's home directory.

Group Account Administration

- Use the `groupadd` command to add a group account:

```
# groupadd [options] group_name
```

- Use the `groupmod` command to modify a group account:

```
# groupmod [options] group_name
```

- Use the `groupdel` command to delete a group account:

```
# groupdel group_name
```

- Use the `gpasswd` command to administer group accounts:

```
# gpasswd [options] group_name
```

- Example: To add a user (`jim`) to a group (`students`):

```
# gpasswd -a jim students
```

- The `groups` command prints the groups to which a user belongs.
- The `newgrp` command changes the real group identification.

User Private Groups

- Each user belongs to a unique group.
 - Eliminates the need for `umask=0022`, which makes groups read-only
 - Allows `umask=0002`, which gives write permission to the group
- Additional steps to implement:
 1. Create a directory to share.
 2. Create a new group.
 3. Add users to this new group.
 4. Change the group ownership for the directory.
 5. Set the `setgid` bit on the directory.



Password Configuration

- Password aging requires users to change their password.
- Use the `chage` command to configure password aging:

```
# chage [options] user_name
```

- Current values are displayed and changed interactively:

```
Minimum Password Age [0]:  
Maximum Password Age [99999]:  
Last Password Change [2011-11-06]:  
Password Expiration Warning [7]:  
Password Inactive [-1]:  
Account Expiration Date [1969-12-31]:
```

- Use the `authconfig` command to configure the password hashing algorithm:

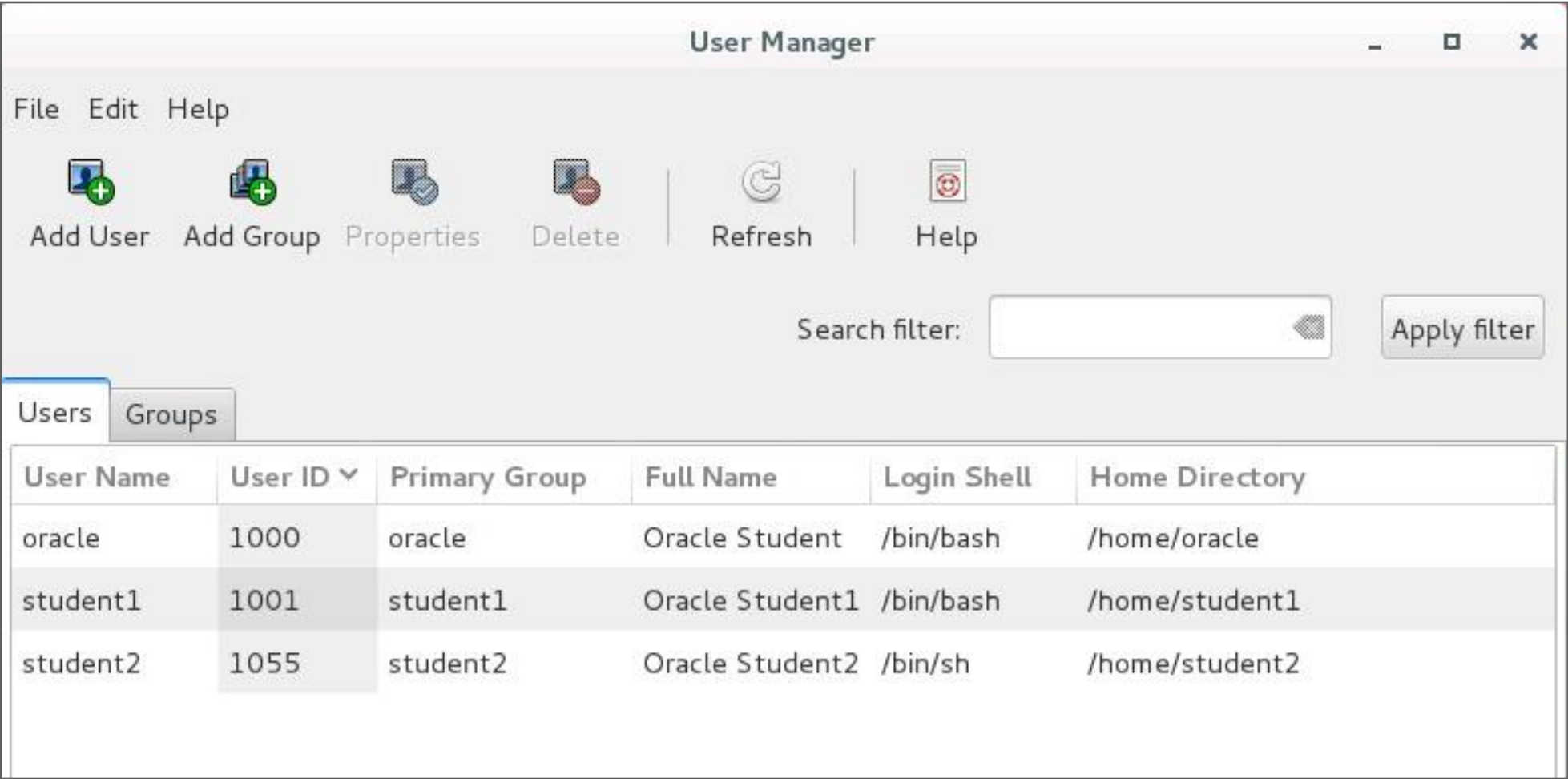
```
# authconfig --passalgo=<algorithm> --update
```

The `/etc/login.defs` File

- The `/etc/login.defs` file provides default user account settings.
- Default values include:
 - The location of user mailboxes
 - Password aging controls
 - Values for automatic UID selection
 - Values for automatic GID selection
 - User home directory creation options
 - The `umask` value
 - The encryption method used to encrypt passwords

The User Manager Tool

The `system-config-users` command starts User Manager.



Restricting Use of the `su` Command

- You can limit access to the `su` command to only those users who are members of the `wheel` group.
- To limit `su` command access to the `oracle` user, add the `oracle` user to the `wheel` group as follows:

```
# usermod -aG wheel oracle
```

- Add the following line to the `/etc/pam.d/su` file to only permit `root` access to members of the `wheel` group:

```
auth    required    pam_wheel.so    use_uid
```

Allowing Use of the `sudo` Command

- `sudo` privileges are configured in the `/etc/sudoers` file.
- The following entry is present in the `/etc/sudoers` file by default:

```
root ALL=(ALL) ALL
```

- The following entry in `/etc/sudoers` allows the `oracle` user to use `sudo` to run administrative commands:

```
oracle ALL=(ALL) ALL
```

- The `oracle` user can now run administrative commands by preceding them with `sudo`. For example:

```
$ sudo useradd new_user  
[sudo] password for oracle:
```

- You are prompted for the `oracle` user password, not the `root` user password.
- By default, commands specified with `sudo` are run as the `root` user.

User/Group Administration in Oracle Cloud Infrastructure

- `opc` is the default user for Oracle Linux image instances.
 - Created when launching an instance
- `opc` is added to the `wheel` group automatically.
- `root` privileges are obtained entirely via `sudo`.
- To obtain sustained `root` privileges:

```
[opc@instance1] ~]$ sudo bash  
[root@instance1 opc]#
```

- `root` privileges are available until `exit` is issued:

```
[root@instance1] opc]# exit  
exit  
[opc@instance1 ~]$
```

User/Group Administration in the Enterprise

- User and group account information is often centralized.
- Centralized information can be retrieved by using:
 - Lightweight Directory Access Protocol (LDAP)
 - Network Information Service (NIS)
- User home directories can also be centralized and accessed remotely.
 - Remote file systems can be auto-mounted.

Quiz



Oracle Linux implements shadow passwords and user private groups.

- a. True
- b. False



Quiz



Which of the following statements are true?

- a. User account information is stored in `/etc/passwd`.
- b. Group account information is stored in `/etc/group`.
- c. Password aging information is stored in `/etc/passwd`.
- d. Default settings for a new user are stored in `/etc/default/useradd` and `/etc/login.defs`.



Summary

In this lesson, you should have learned how to:

- Explain user and group implementation
- Describe user and group configuration files
- Configure users and groups by using command-line utilities
- Implement user private groups (UPG)
- Configure password aging and the hashing algorithm
- Use the User Manager GUI tool
- Manage the use of `su` and `sudo` commands
- Describe user and group administration in Oracle Cloud Infrastructure
- Describe user and group implementation in the enterprise



Practice 11: Overview

This practice covers the following topics:

- Administering user accounts
- Administering group accounts
- Implementing user private groups
- Configuring password aging
- Using the User Manager GUI
- Restricting the use of the `su` command
- Allowing the use of the `sudo` command